



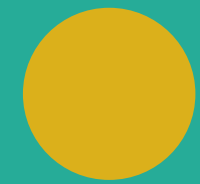
Machine Learning Model on Refugee Population Forecasting and Humanitarian Resource Planning for Kenya



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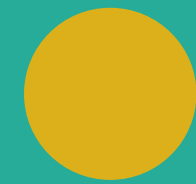
1. Project Overview



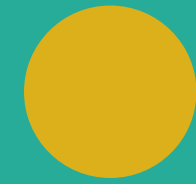
2. Business problem and Objectives



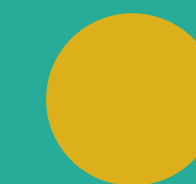
3. Data Understanding



4. Business Insights and Visualisation



5. Modelling



6. Recommendations and Conclusion

PROJECT OVERVIEW



Over 100 million people forcibly displaced worldwide — a record high in modern history.



Kenya hosts one of the largest refugee populations in Africa



Average displacement lasts over 10 years, making long-term support essential for refugees.

Business Problem



Humanitarian agencies often face challenges in forecasting refugee population changes, resulting in resource shortages or inefficient allocation of aid.

This project aims to develop a predictive analytics

and deep learning system capable of forecasting refugee population trends across demographic groups and supporting evidence-based humanitarian planning.



BUSINESS OBJECTIVES

- ➔ Refugees face critical obstacles in accessing shelter, healthcare, education, and employment. These barriers create cycles of vulnerability, leaving millions without the resources needed to rebuild their lives.
- ➔ The project aimed to help humanitarian organizations:
 - ✓ Anticipate future refugee populations
 - ✓ Improve planning accuracy
 - ✓ Allocate resources efficiently
 - ✓ Support evidence-based policy decisions
 - ✓ Reduce emergency response delays



→ Our analysis focused on answering three key questions.

1. How have refugee and asylum seeker populations in Kenya changed over time?
2. Which demographic and time-related factors influence refugee population trends?
3. How can predicted refugee populations improve humanitarian resource planning?





END TO END PIPELINE

 Explore Data Analyzing 25 years of UN refugee logs

 Train Models Benchmarking Classical ML vs. Deep Learning

 Forecast Demand Predicting segment-level population changes

 Feature Engineering and Model Development

 Deploy App Interactive cloud interface for decision-makers



DATA UNDERSTANDING

Dataset Dimensions

The dataset comprises 27,664 observations with 14 distinct variables for analysis.

25 years of refugee data (2001–2025)

Variable Count

Fourteen variables offer diverse perspectives on the refugee population characteristics

Population was the target variable



VISUALIZATION

Refugee Population Trend Over Time



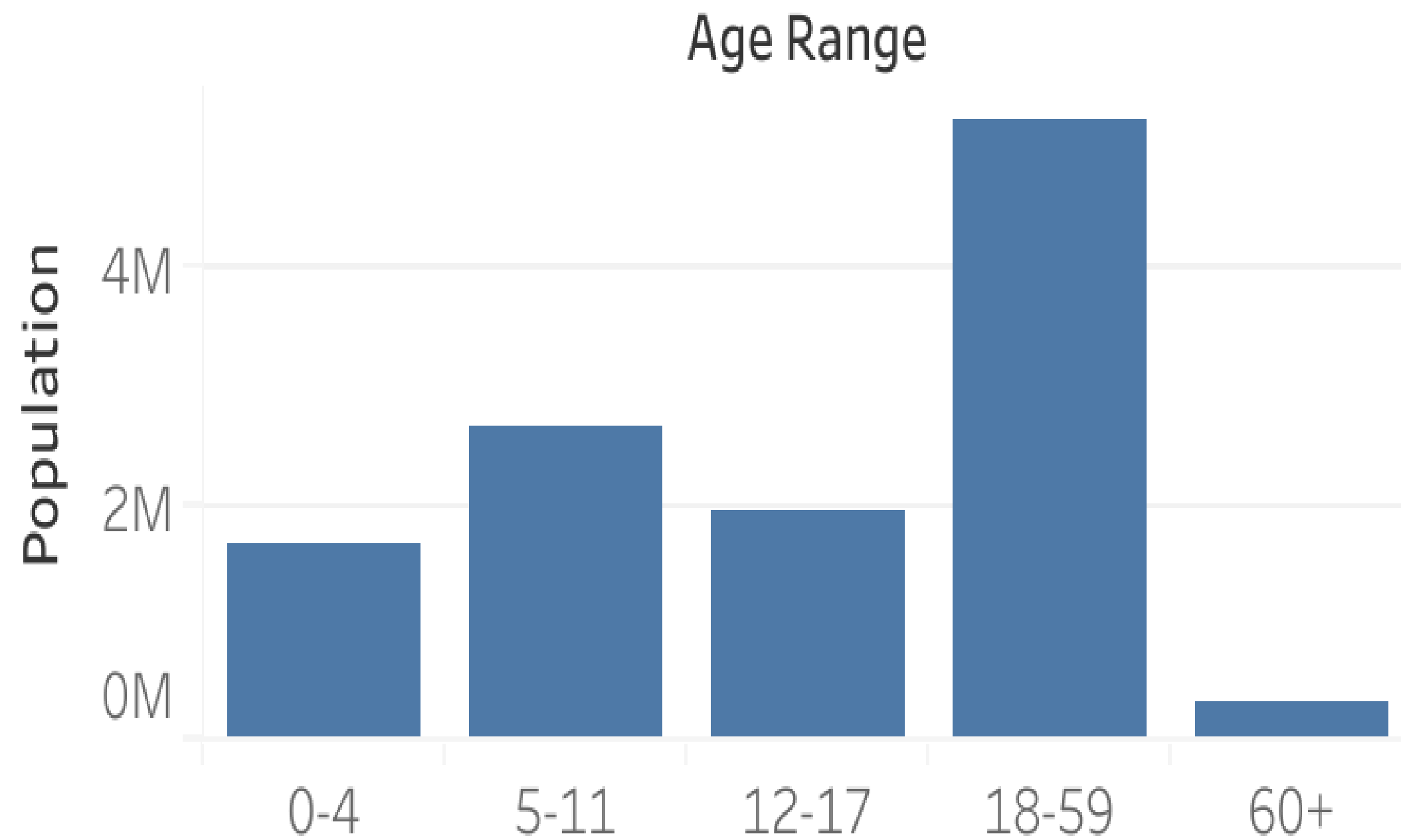
Insight

The refugee population generally increased over time, peaking in 2025 after a sharp rise from 2023.



VISUALIZATION

Refugee Population Across Age Ranges



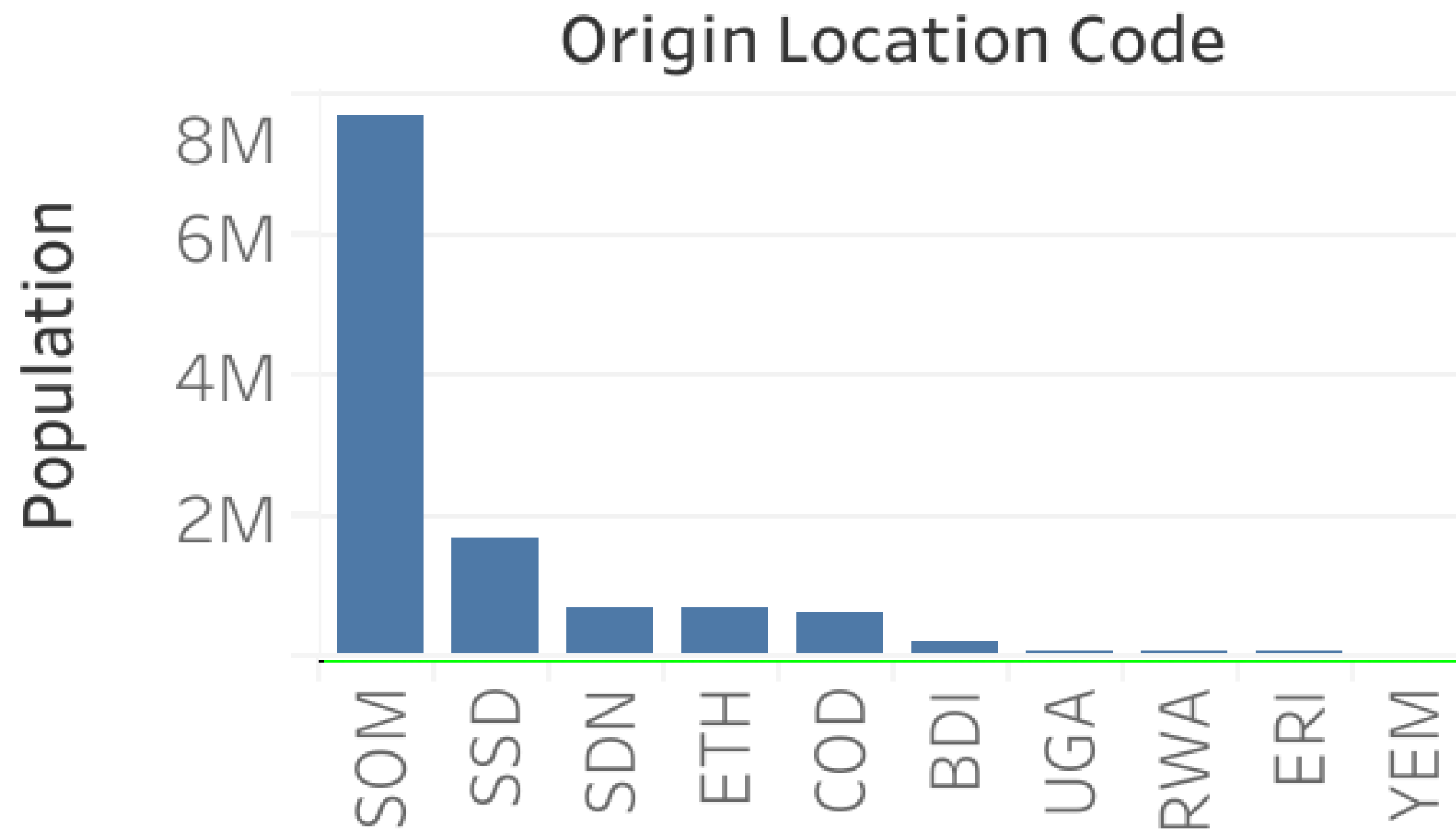
Insight

The 18-59 age group dominates for both genders while the 60+ group is the smallest.



VISUALIZATION

Top 10 Refugee Population By Origin Location

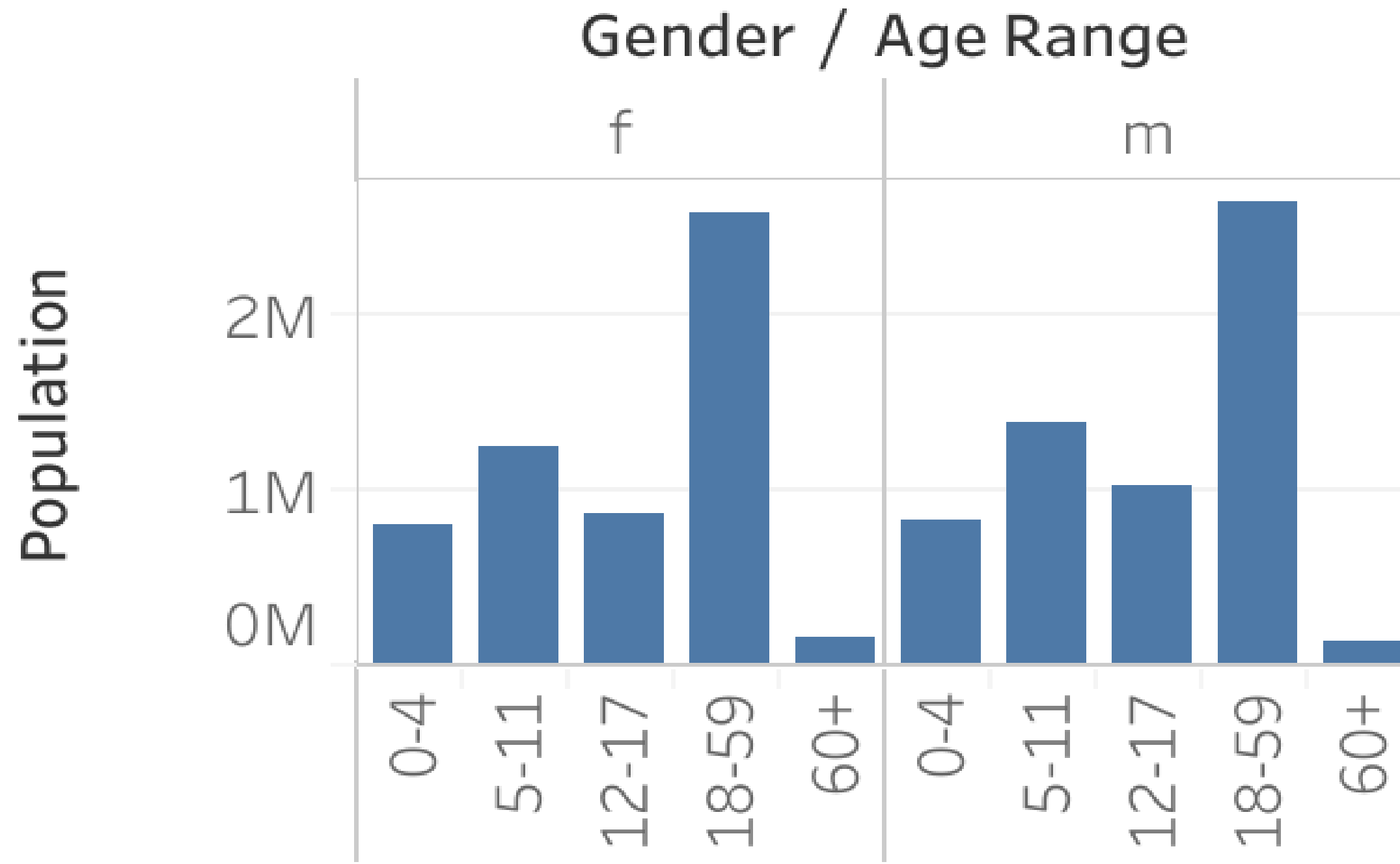


Insight

Somalia is the dominant country of origin, contributing far more refugees than any other country, with South Sudan a distant second.

VISUALIZATION

Population by Age Range and Gender

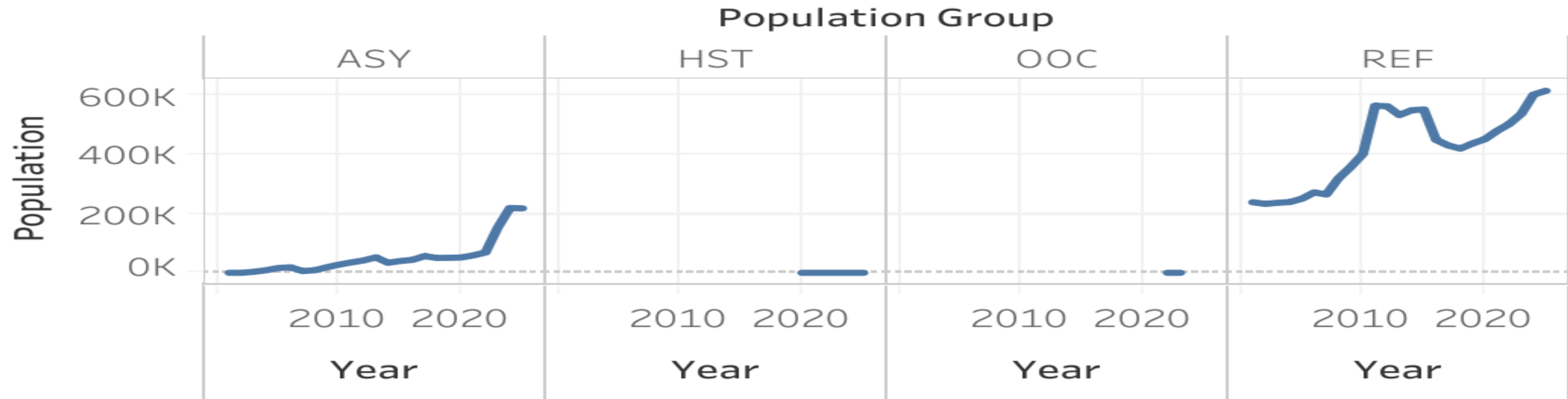


Insight

The 18-59 age group dominates for both genders while the 60+ group is the smallest across both genders.

VISUALIZATION

Population by Time and Population Group



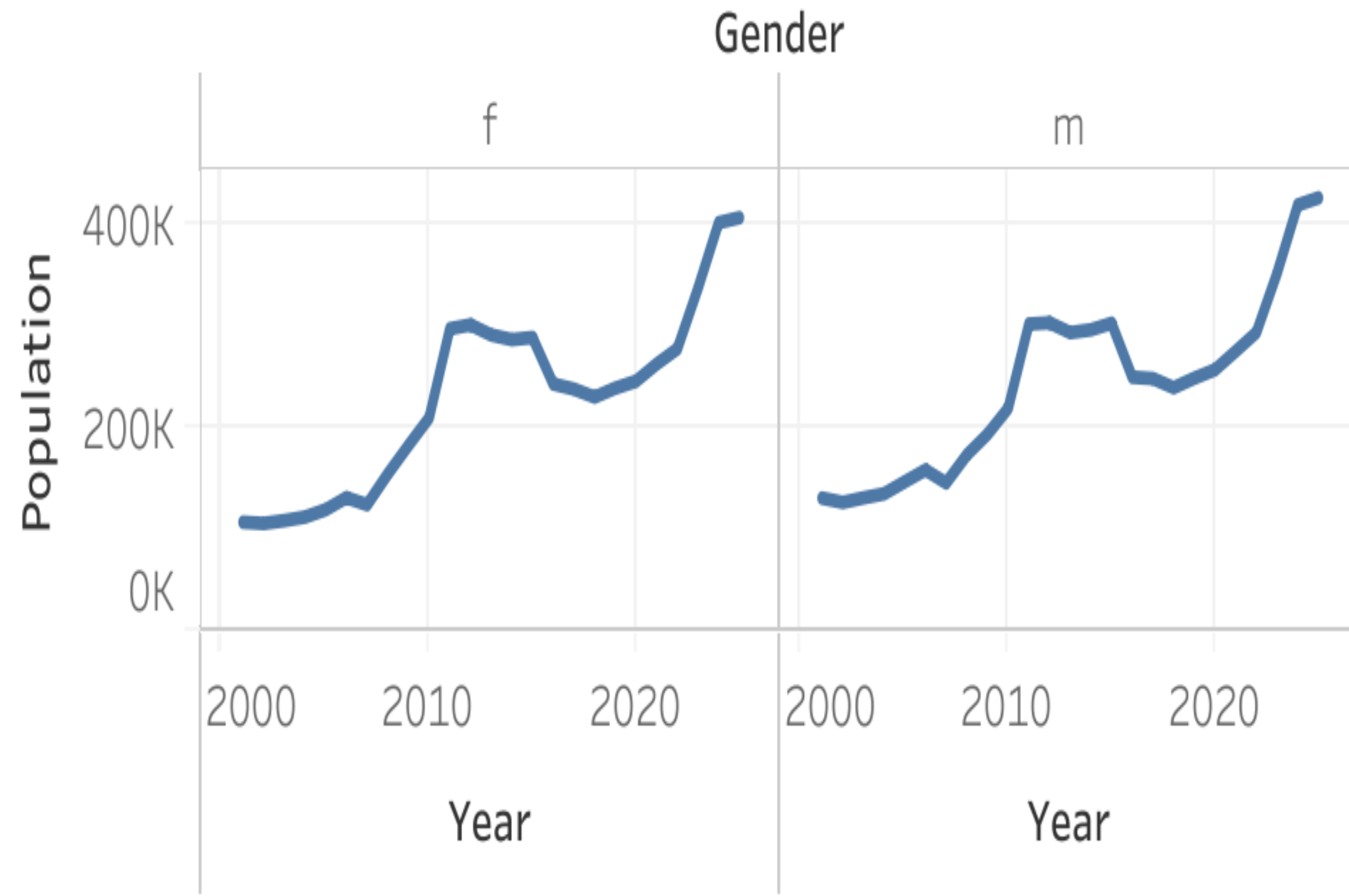
Insight

Refugees make the largest population group and shows the strongest growth.



VISUALIZATION

Population By Gender Over Time



Insight

Male and female populations show similar trends and increase significantly over time, indicating a possible increase in the near future..



MODELLING



Several forecasting approaches were tested.
Each model learned from historical refugee trends and attempted to predict future populations.
The models were compared based on:

- Prediction accuracy
- Reliability
- Consistency

The best-performing model was selected for deployment.



MODEL COMPARISON

	Model	MAE	RMSE	R ²
4	FT Transformer	0.214968	0.252495	0.924394
1	Decision Tree	484.926760	2282.006131	0.881252
2	Random Forest	466.627194	2285.698510	0.880868
3	XGBoost	548.369995	2373.535864	0.871536
0	Linear Regression	2174.563168	5005.000625	0.428785

Insight

The best project(FT Transformer) based on R2 was selected for tuning and deployment



MODEL SUCCESS METRICS

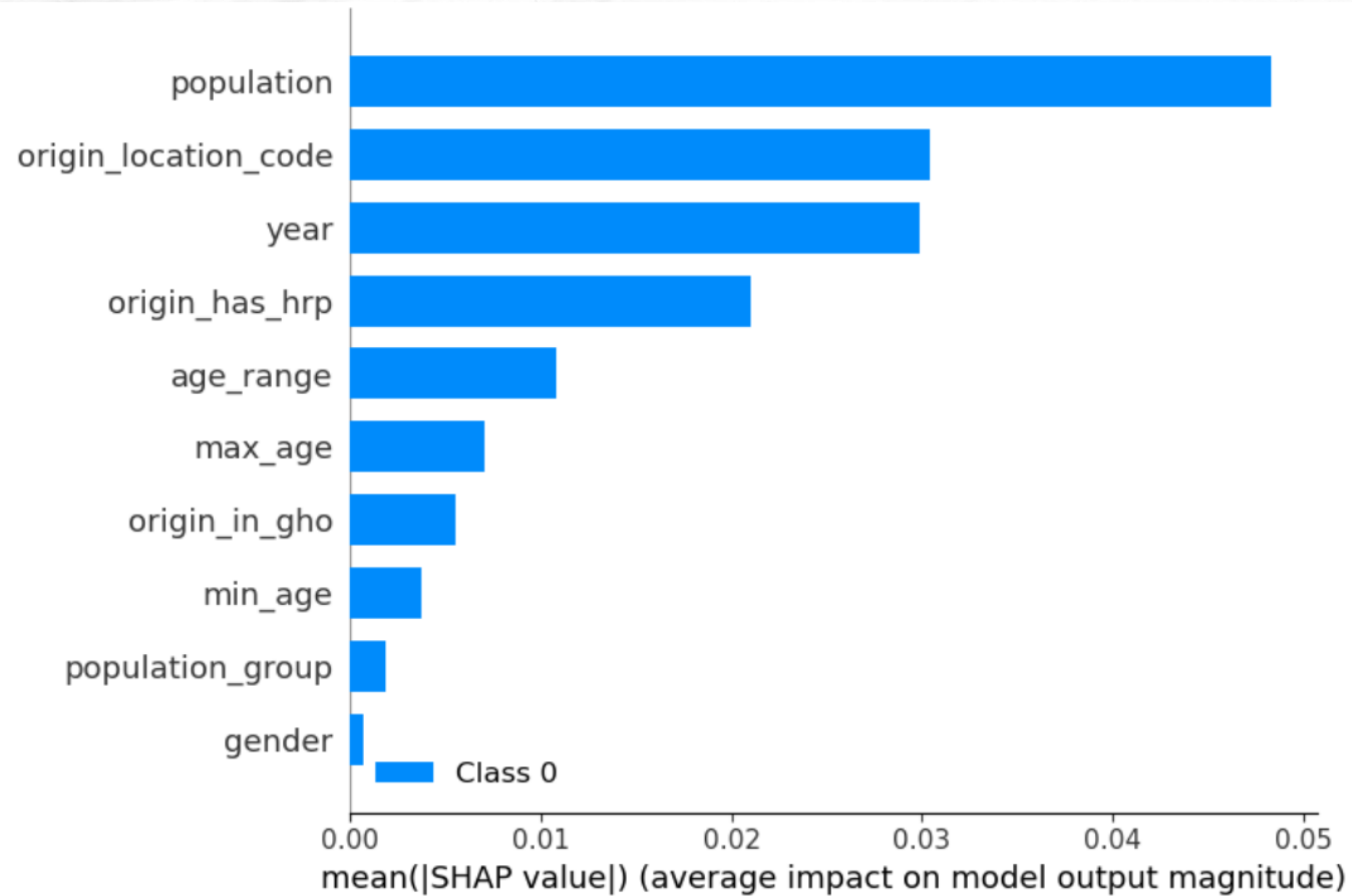
The Ft transformer model achieved an R^2 of 92.4% of the variation in refugee population, making it highly accurate.

It Recorded the lowest MAE and RMSE, indicating highly accurate predictions.

The model Provides dependable forecasts that support better humanitarian planning and efficient allocation of resources.



MODEL INTERPRETABILITY



Insight

Origin location, year and origin has hrp were the factors that affect population the most.



MODEL DEPLOYMENT

Model Metadata

- Architecture: FT-Transformer (Feature Tokenizer Transformer)
- Framework: PyTorch (Deep Learning)
- Prediction Target: Refugee Cohort Population Size
- Horizon: 2026–2030

Model Evaluation

Showing Model Validation Baseline metrics.

R² Score (Variance Explained) ⓘ

0.912

Mean Absolute Error (MAE) ⓘ

142.5 indivi...

Root Mean Squared Error (RMSE) ⓘ

184.2 indivi...

AI-Powered Refugee Population Forecasting System

Project Objective

To provide humanitarian organizations with a proactive tool for estimating localized refugee demographic trends in Kenya and automatically calculating downstream resource allocation.

This dashboard uses a trained **Feature Tokenizer Transformer (FT-Transformer)** deep learning model to forecast refugee population trends in Kenya and project crucial logistics requirements like daily water, food distribution, and emergency housing.

✓ FT-Transformer model loaded successfully. Ready to generate refugee population forecasts.

> ☐ User Manual & Quick Terminology Glossary

Demographic Parameters

Country of Origin ⓘ

AFG

Population Group Type ⓘ

ASY

Gender Cohort ⓘ

f

Age Range ⓘ

0-4

Model Inference & Resource Forecasting

Current Baseline Population (Historical) ⓘ

5000

 Generate Forecast

Insight

The model was deployed on Streamlit

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0-4

Forecasting Timeline

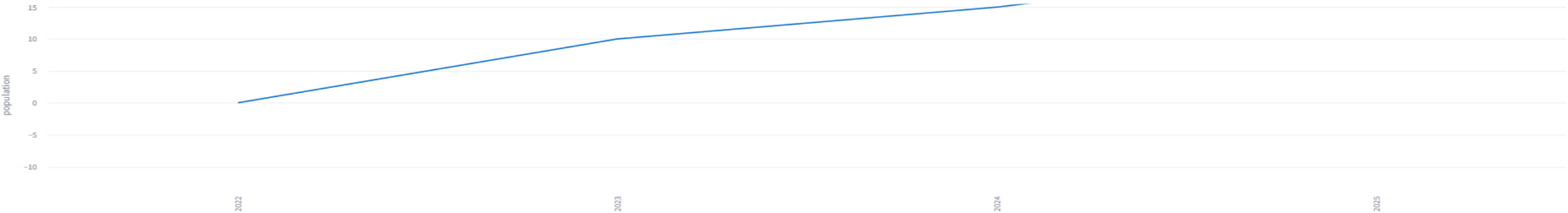
Target Forecast Year

2026

Geopolitical Indicators

- Origin country has active Humanitarian Response Plan (HRP)
- Included in Global Humanitarian Overview (GHO)
- Kenya has active Humanitarian Response Plan (HRP)
- Kenya included in Global Humanitarian Overview (GHO)

Historical Population Trend Analysis (2022 - 2025)



Showing recorded historical population timeline from 2022 to 2025 for f cohorts aged 0-4 from AFG.

Insight

The model was deployed on Streamlit

INSIGHTS

How Have Refugee and Asylum Seeker Populations Changed Over Time?

- Refugee populations have shown substantial changes over the past 25 years.
- Some years experienced rapid increases due to regional instability.
- Population growth is not constant and requires continuous monitoring.



INSIGHTS

Which Demographic and Temporal Factors Influence Refugee Population Trends?

The most influential factors include:

- Previous population trends
- Age group
- Gender
- Refugee category
- Country of origin
- Reporting period



INSIGHTS

How Can Population Forecasts Improve Humanitarian Planning?

Forecasts help organizations estimate future demand for:

- Food assistance
- Shelter
- Healthcare
- Schools
- Water and sanitation
- Humanitarian staffing



RECOMMENDATIONS

Based on the analysis, we recommend:

1. Use predictive forecasting for annual humanitarian planning.
2. Integrate real-time information such as:
 - Conflict events
 - Weather conditions
 - Climate risks
 - Border movements
3. Retrain the forecasting model regularly as new data becomes available.
4. Develop interactive dashboards for decision makers.
5. Use forecasts to guide national resource allocation and policy planning.



CONCLUSION

- ✓ Better forecasts lead to better allocation of food, shelter, healthcare, education, and funding.
- ✓ Machine learning can become a valuable decision-support tool for government and humanitarian agencies.



THANK YOU

